Northwind Traders

Optimizing Revenue: Average Ticket & Customer Retention

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Resume

This report provides a comprehensive analysis of revenue dynamics, customer behavior, and product performance, with a focus on identifying key drivers of average ticket growth and customer churn. By combining product-level insights, revenue concentration patterns, and customer lifecycle analysis, the study highlights structural inefficiencies in value generation and retention.

The findings reveal that revenue growth is currently driven by order volume rather than ticket expansion, while customer value is strongly linked to purchase frequency. Additionally, churn is primarily concentrated in the transition from moderately active to loyal customers, indicating a critical gap in customer progression. Together, these insights uncover clear opportunities to optimize both average ticket and retention through targeted, data-driven strategies.

Summary

Revenue Overview & Growth Dynamics
- Revenue Trend Analysis
- Order Volume Evolution
- Average Ticket Behavior

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- Cross-sell and Upsell Opportunities

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- Top Product Contribution
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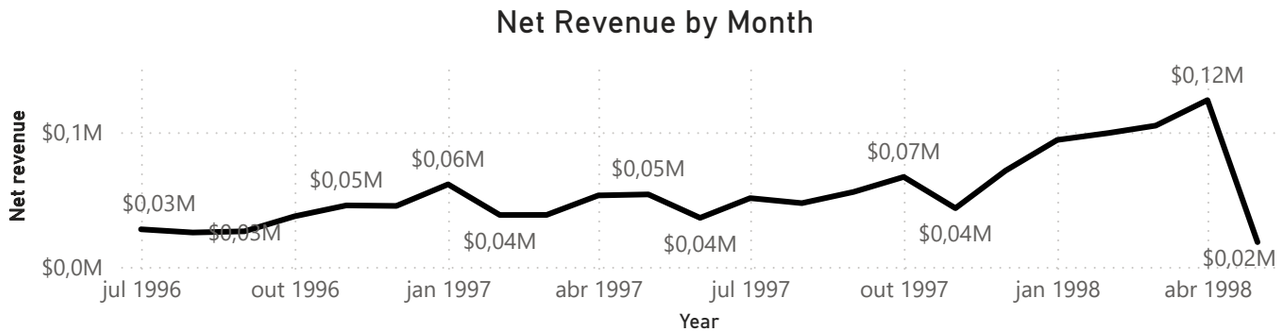
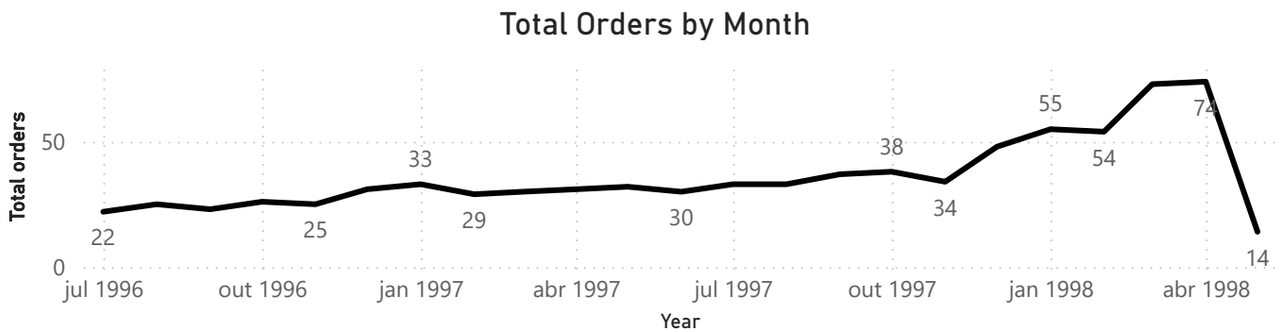
- Strategic Actions

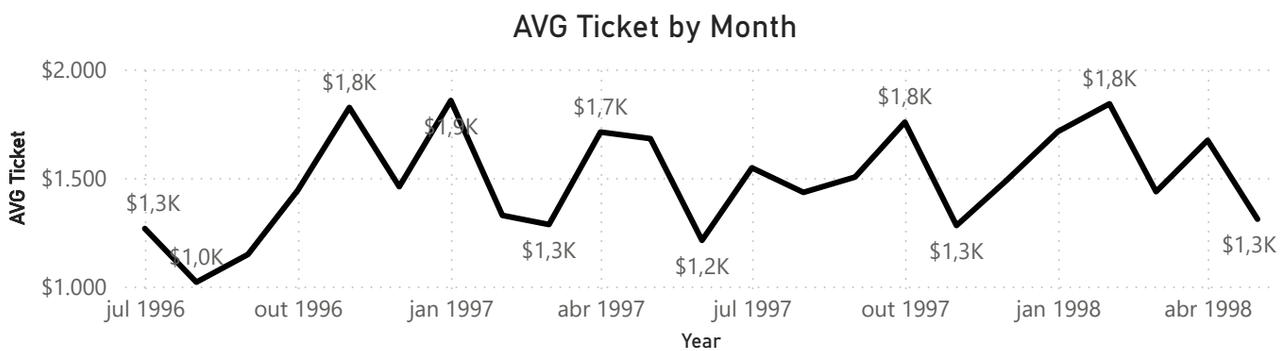
Github Respository

- *The GitHub repository contains all files related to data modeling, containerization, and database structuring, as well as the .pbix file for this dashboard.*

Net Revenue	Orders	AVG Ticket	Total Customers	Date
$1,27M	830	$1.525	89	04/07/1996 – 06/05/1998







Revenue Trend & Growth Dynamics

Revenue shows a consistent growth trend over the period, increasing from approximately $0.03M to peaks above $0.12M. A stronger acceleration is observed from late 1997 onwards, indicating a potential gain in operational traction or demand expansion. The sharp decline in the final data point suggests a one-off effect (seasonality or incomplete data) rather than a structural reversal.

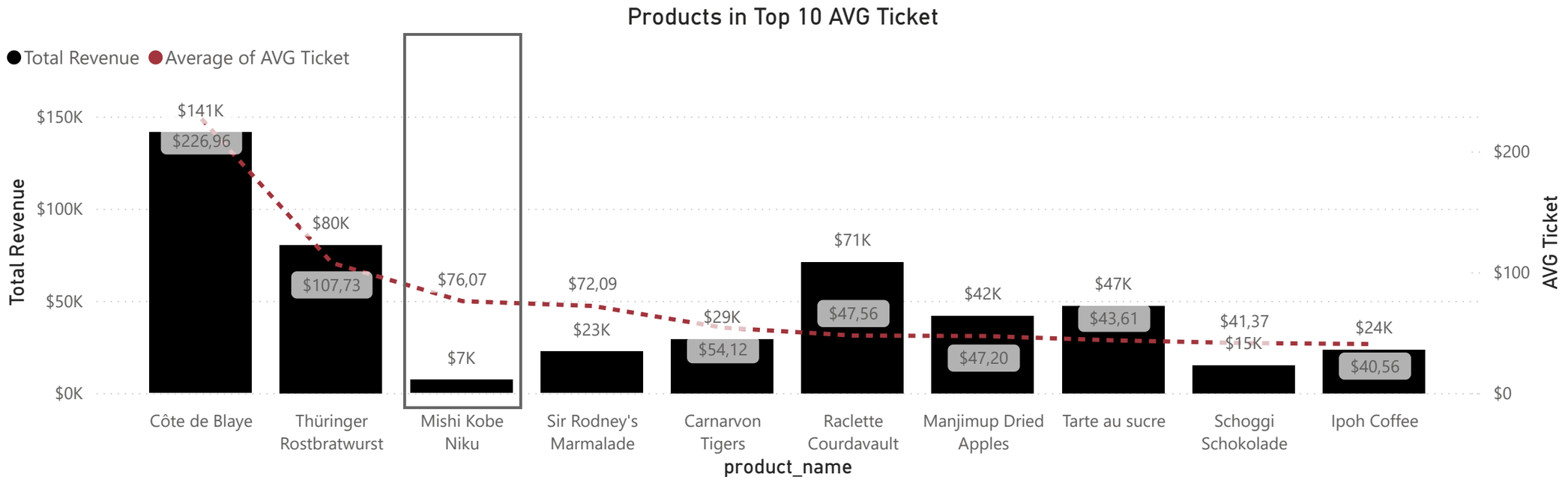
Order Volume Evolution

Order volume follows the same upward trend as revenue, growing from around 22 to over 70 orders per month. Growth is relatively stable, with minor fluctuations, indicating a gradual expansion of the customer base or increased recurrence. Similar to revenue, the drop in the final period reinforces the likelihood of an outlier or partial-period effect

Average Ticket Behavior

Average ticket remains relatively stable within the $1.2K to $1.8K range, with peaks close to $2.0K, indicating consistency in transaction value. There is no clear structural upward trend in ticket size, suggesting that revenue growth is being primarily driven by volume (number of orders). Ticket fluctuations indicate tactical optimization opportunities, but not yet a fully leveraged growth driver

Value vs Volume Misalignment

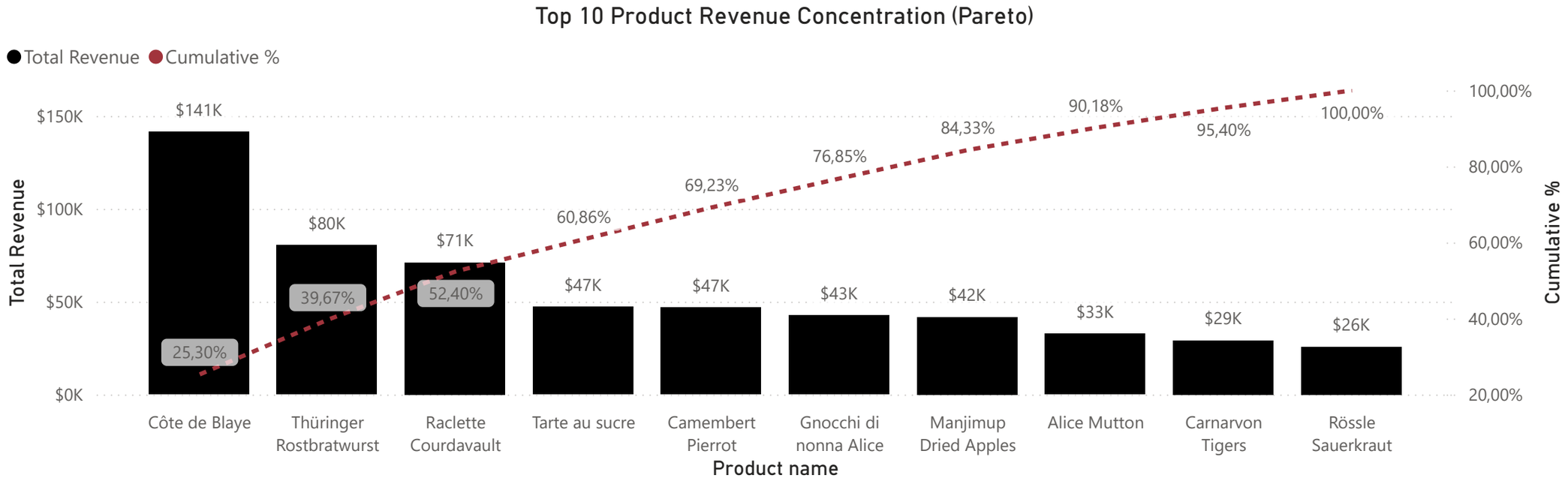


Cross-sell and Upsell opportunities

The analysis reveals products with high average ticket values but low revenue contribution, indicating a misalignment between value generated per transaction and sales volume. These items represent a strong lever for increasing the overall average ticket if properly activated.

To capture this opportunity, the business can implement targeted **cross-sell strategies**, associating these high-value items with high-volume products, as well as **upsell initiatives**, promoting premium alternatives, larger formats, or upgraded versions at key decision points. Additionally, improving product positioning and visibility can help increase adoption and drive higher-value transactions.

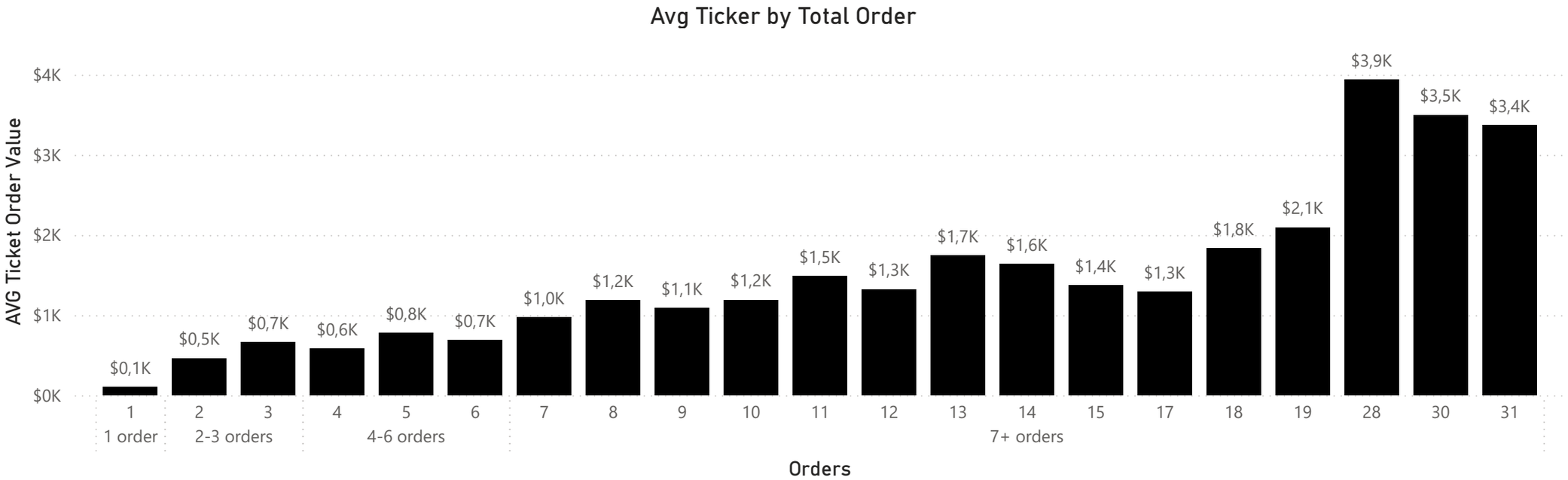
Revenue Concentration



Bundle opportunity

The analysis reveals that the top four products account for approximately **61% of total revenue**, indicating a high concentration and dependence on a limited set of items. This dynamic presents a clear opportunity to leverage these high-performing products as anchors in bundling strategies. By combining them with complementary or lower-turnover items, it is possible to not only diversify product exposure but also drive higher-value transactions, ultimately increasing the overall average ticket.

Impact of Customer Frequency on Average Ticket

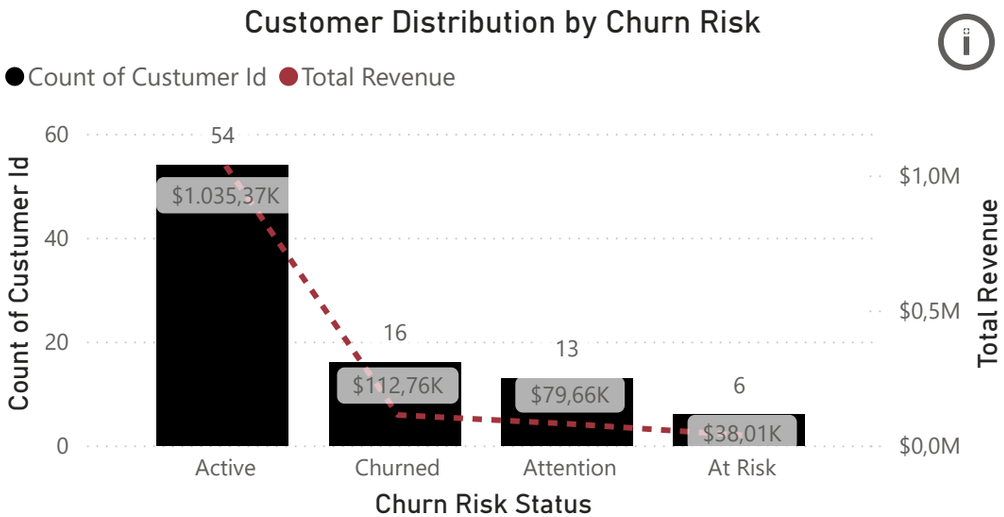


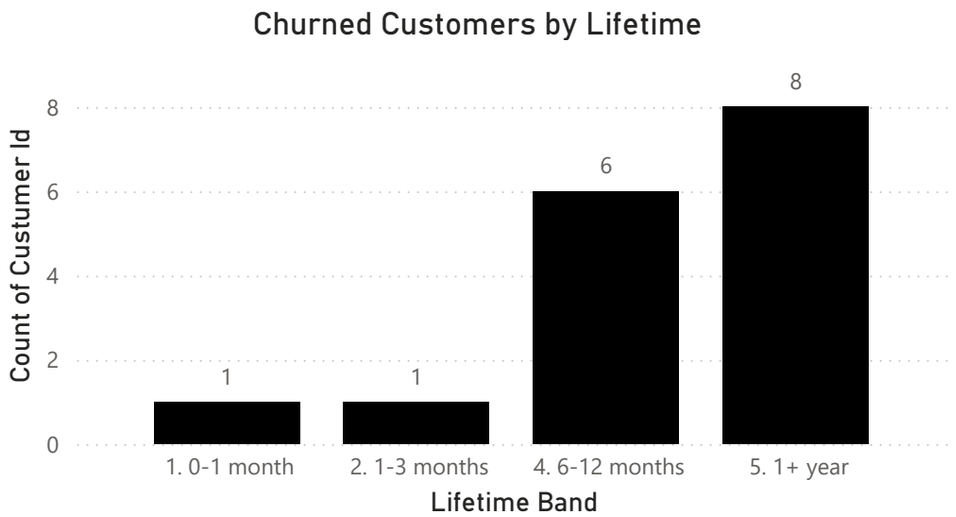
Higher purchase frequency significantly increases average ticket, highlighting retention as the key lever for sustainable revenue growth
A strong revenue concentration is observed, where **a small number of products account for approximately 61% of the total**, while most orders remain within the $0.5M to $1.5M ticket range. At the same time, behavioral analysis shows that more recurrent customers significantly increase their average ticket, evolving from approximately $0.1M in a single purchase to up to $3.9M among the most frequent ones.

This pattern highlights that revenue growth is not driven solely by new customer acquisition, but primarily by customer retention and progression throughout the purchase lifecycle, where higher frequency is directly associated with greater value generation.
Therefore, the current revenue concentration, combined with the uneven ticket distribution, reveals a clear growth opportunity through expanding the recurring customer base and encouraging customers to move toward higher consumption tiers.

Churn & Retention

6,74%	3,0%	8,91%	18,0%
Customers at Risk %	Revenue at Risk %	Lost Revenue %	Churned Customers %

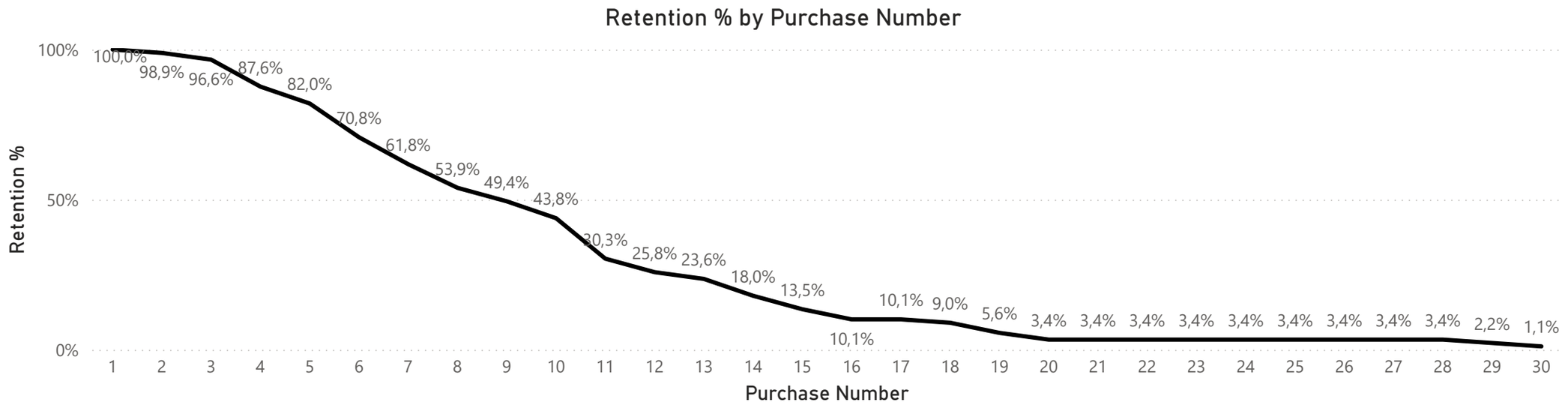




The analysis shows that, although **churn represents around 18% of the customer base**, its direct impact on revenue remains relatively limited, as it is primarily concentrated among lower-value customers. However, a more critical pattern emerges when observing that the most relevant losses occur among more mature customers, particularly after 6 months of engagement. This indicates a structural retention challenge, suggesting that the issue lies not in initial acquisition, but in sustaining customer engagement and value over time.

Given this, it becomes essential to implement lifecycle-based retention strategies, with targeted actions tailored to each stage of the customer journey, especially during high-risk periods such as 6 to 12 months and beyond one year. Additionally, proactive monitoring mechanisms, such as identifying declines in purchase frequency and average ticket, can help anticipate churn. Complementing this, loyalty programs and progressive incentives can strengthen relationships with recurring customers. Finally, focusing on the “at-risk” segment, which is small but highly actionable, represents a high-impact opportunity to preserve revenue and maximize long-term customer value.

Churn & Retention



The highest drop-off occurs between the 5th and 7th purchases, highlighting a critical gap in converting repeat customers into loyal ones.
The repurchase funnel shows that customer drop-off is most pronounced between the fifth and seventh purchases, indicating a critical gap in the transition from early repeat buyers to loyal customers. While initial engagement is strong, many customers fail to sustain consistent purchasing behavior beyond this stage.

This pattern suggests that churn is primarily driven by the inability to convert moderately active customers into habitual ones. Strengthening engagement during this phase through targeted incentives, milestone-based rewards, and personalized cross-sell strategies is key to improving retention, reducing churn, and maximizing long-term customer value.

Actionable Insights (Ticket + Churn)

Quick Wins (high impact / low effort)

- **Implement product bundling with top performers:**
  Combine high-revenue products (responsible for ~61% of revenue) with lower-turnover items to increase basket size and drive higher average ticket.

- **Trigger post-purchase campaigns for early-stage customers (2nd–5th purchase):**
  Use targeted discounts, reminders, and personalized offers to accelerate progression and reduce early-stage churn.

- **Promote high-ticket, low-volume products via cross-sell:**
  Integrate premium items into the purchase journey of high-volume products to immediately increase transaction value.

- **Introduce threshold-based incentives (e.g., spend X, get Y):**
  Encourage customers to increase order value through minimum spend benefits (free shipping, discounts, add-ons).

Actionable Insights (Ticket + Churn)

Strategic Actions (medium/long-term)

- **Develop a structured customer progression strategy (lifecycle-based)**
  Create tailored actions for each stage (early, mid, loyal), focusing especially on customers between the 4th and 7th purchases where churn peaks
- **Build a “habit formation” program**
  Implement milestone-based rewards (e.g., benefits on 3rd, 5th, 7th purchases) to convert repeat buyers into loyal customers
- **Optimize product mix and value ladder**
  Address the gap between low-ticket and premium products by introducing mid-tier offerings and better positioning high-value items
- **Implement churn prediction signals**
  Monitor early indicators such as decline in purchase frequency, lower ticket size, and increased time between orders to trigger proactive retention actions
- **Strengthen retention as a revenue strategy**
  Shift focus from acquisition-driven growth to retention-driven growth, leveraging the strong correlation between frequency and average ticket

Github Repository

Strategic Actions (medium/long-term)

The project was developed with an analytics engineering approach, organizing Northwind source data into structured analytical layers (raw, staging, intermediate, and marts) to support performance indicators, sales analysis, and churn monitoring. The solution was built using PostgreSQL, dbt, Python, Docker Compose, and Power BI, combining data ingestion, best-practice data modeling, and analytical consumption for both operational and executive reporting.

The complete repository, including documentation, operational runbook, data dictionary, and project artifacts, is available at:
https://github.com/nicklucarangel/northwind.git